CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 2: Generic Data Structures with Java Collections

Course: Advanced Java Programming Laboratory [23CS3504]
Category: Experiments / Practicals | Experiment #2 | Academic Year: 2026-27

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1. Overview & Objective:

Develop type-safe custom Stack and Queue data structures utilizing Java Generics.

2. Practical Tasks & Verification Steps:

1. Create generic class CustomStack<T> with push , pop , and peek methods using dynamic array resizing.
2. Implement custom Queue<T> utilizing doubly linked nodes with O 1 enqueue and dequeue operations.
3. Demonstrate bounded generic wildcards <? extends Number>, <? super Integer> .
4. Benchmark custom collection performance against java.util.ArrayDeque across 100,000 operations.

Submission Guidelines & Verification:

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